

ISRC Robotics Event: Water Boat Racing Rules, Regulations, and Judging Criteria

Rules and Regulations

1. Eligibility

- ❑ **Age Groups:**
 - **Group 1:** 10-14 years
 - **Group 2:** 15-18 years
- ❑ **Team Size:** Teams must consist of 2- 5 members.

2. Event Structure

- ❑ **Rounds:**
 - **First Round:** All teams participate .
 - **Second Round:** All teams participate .
 - **Final Round:** Top teams from the second round compete in a final race, featuring the most complex course with additional obstacles and turns.
- ❑ **Attempts:** Each team will be given two attempts per round. The best attempt (shortest time with the fewest penalties) will be considered for scoring.

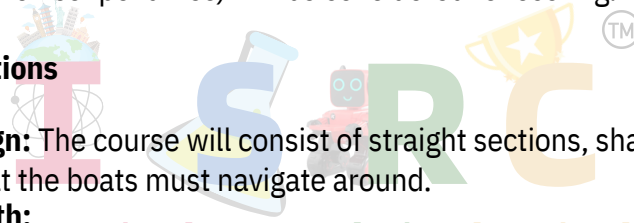
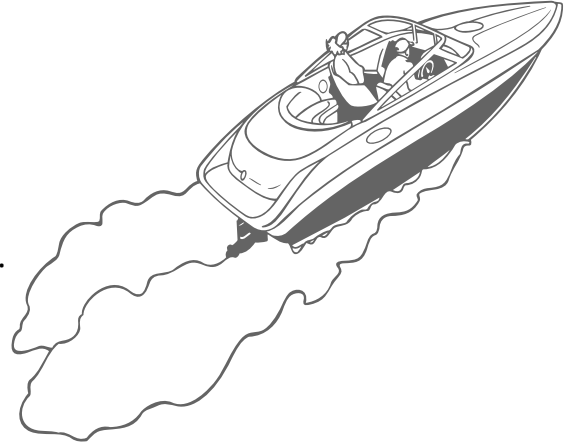
3. Course Specifications

- ❑ **Course Design:** The course will consist of straight sections, sharp turns, and floating obstacles that the boats must navigate around.
- ❑ **Course Length:**
 - **Group 1 (10-14 years):** 10 meters with moderate difficulty.
 - **Group 2 (15-18 years):** 15 meters with increased difficulty, including tighter turns and more obstacles.
- ❑ **Water Depth:** The water depth will be consistent across both groups at minimum 30 cm to ensure stability for all boat designs.
- ❑ **Course Width:** The course will be 1 meter wide to allow for smooth navigation while challenging participants to design stable and maneuverable boats.

4. Boat Requirements

- ❑ **Autonomous or Remote-Controlled:** Boats may either operate autonomously or be controlled remotely.
- ❑ **Size Restrictions:** Boats must fit within a 30x30 cm footprint.
- ❑ **Power Supply:** Boats must be powered by batteries or solar energy. No fuel-powered boats are allowed.
- ❑ **Material Usage:** Boats should be constructed from lightweight, water-resistant materials. Teams are encouraged to use eco-friendly materials where possible.

5. Competition Format



- ☐ **Time Trials:** The boat's performance is timed, and the fastest time to complete the course with the fewest penalties is recorded.
- ☐ **Obstacle Navigation:** Boats must navigate the course without colliding with obstacles or getting stuck. Deviating from the course will incur penalties.
- ☐ **Restarts:** Teams are allowed to restart their boat if it fails to navigate the course, but each restart will incur a time penalty.

6. Scoring and Penalties

- ☐ **Time Scoring:** The boat with the shortest time and fewest penalties in each round will receive the highest score.
- ☐ **Penalty Points:**
 - o **Colliding with an obstacle:** +5 seconds
 - o **Getting stuck:** +10 seconds
 - o **Restarting the boat:** +15 seconds
- ☐ **Final Scoring:** The best score from the two attempts in each round will be considered for advancing to the next round and for final scoring.

7. Safety Regulations

- ☐ **Boat Safety:** Boats must be designed safely with no sharp edges or hazardous materials.
- ☐ **Course Safety:** The course area must be kept clear of all participants and spectators during the runs.
- ☐ **Emergency Stop:** Each boat must have an accessible emergency stop mechanism.

Judging Criteria for Water Boat Racing Event

International Stem & Robotics Championship

Scoring Out of 1000 Points:

1. Innovation and Originality (200 points)

- ☐ **Creative Navigation Techniques (80 points):** Evaluate the originality of the boat's design and approach to navigating the course, including the use of innovative sensors or control systems.
- ☐ **Advanced Problem-Solving (60 points):** Assess the boat's ability to handle unexpected challenges or deviations in the course.
- ☐ **Novelty in Design (60 points):** Consider the uniqueness of the boat's physical design and its potential to introduce new concepts in watercraft engineering.

2. Technical Mastery (300 points)

- ☐ **Precision in Course Navigation (120 points):** Evaluate the accuracy with which the boat navigates the course, maintaining consistent direction and speed.
- ☐ **Stability and Control (100 points):** Assess the effectiveness of the boat's design in maintaining stability and control, especially in rough or obstacle-filled waters.
- ☐ **Durability and Reliability (80 points):** Evaluate the boat's ability to function reliably under various conditions within the course.

3. Practical Applicability and Global Impact (150 points)

- **Real-World Applications (70 points):** Consider the potential of the boat's design to be applied in real-world scenarios, such as search and rescue operations or environmental monitoring.
- **Environmental Considerations (50 points):** Evaluate the sustainability of the boat, including energy efficiency and material usage.
- **Scalability (30 points):** Assess the potential for the boat's design to be scaled for larger or more complex applications.

4. Presentation Excellence (200 points)

- **Technical Explanation (100 points):** Evaluate the team's ability to explain the technical aspects of their boat, including design choices, material selection, and control systems.
- **Demonstration Quality (70 points):** Assess the effectiveness of the live demonstration, including the boat's performance and the clarity of the team's presentation.
- **Team Communication (30 points):** Consider how well the team collaborates and communicates their ideas during the presentation.

5. Judges' Challenge (100 points)

- **Real-Time Problem Solving (50 points):** Present a real-time challenge to the teams, such as modifying the course or introducing a new obstacle, and evaluate their ability to adapt.
- **Cross-Disciplinary Collaboration (30 points):** Assess how well the team integrates knowledge from different STEM fields, such as engineering, programming, and design.
- **Time-Constrained Innovation (20 points):** Evaluate the team's ability to innovate under time constraints during the challenge.

6. Boat Design and Functionality (50 points)

- **Mechanical Design (20 points):** Assess the structural design, including the materials used, compactness, and overall build quality of the boat.
- **Programming and Control (20 points):** Evaluate the boat's programming logic, control algorithms, and efficiency in executing tasks.
- **Adaptability (10 points):** Consider the boat's ability to adapt to different course conditions, including speed adjustments and obstacle avoidance.